

Homework 1: Concise Answers

Intro1

Rational action maximizes expected utility given the available information I :

$$a^* \in \arg \max_a \mathbb{E}[U \mid I, a].$$

Rationality concerns the decision method; selfishness concerns the goal.

$$U = U_{\text{self}} \quad (\text{selfish}), \quad U = \alpha U_{\text{self}} + \beta U_{\text{others}} \quad (\text{may be altruistic}).$$

Example: compare price and quality, then choose the best expected value. A rational choice may still have a bad outcome under uncertainty.

Intro2

Example: exhaustive game-tree search under a short time limit. For branching factor b and search depth d ,

$$N_{\text{leaves}} = b^d, \quad N_{\text{nodes}} = \sum_{k=0}^d b^k = \frac{b^{d+1} - 1}{b - 1} \quad (b > 1).$$

Humans have limited memory and calculation speed. Computers can systematically enumerate and evaluate many moves; AI need not imitate human thinking.

Intro3

LLM: Codex; original discussion: October 8, 2026.

Prompt: Why did multilayer learning take years despite XOR needing only one extra hidden layer?

x_1	x_2	z_1 (OR)	z_2 (AND)	y (XOR)
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

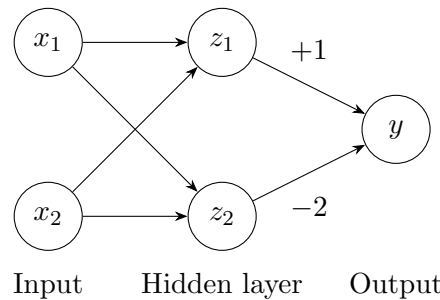
For a single linear threshold, XOR would require

$$b < 0, \quad w_1 + b \geq 0, \quad w_2 + b \geq 0, \quad w_1 + w_2 + b < 0.$$

But $w_1 + w_2 + b \geq -b > 0$: contradiction. Thus XOR is not linearly separable.

Let $H(t) = \mathbf{1}[t \geq 0]$. A nonlinear hidden layer solves XOR:

$$\begin{aligned} z_1 &= H(x_1 + x_2 - 0.5), & z_2 &= H(x_1 + x_2 - 1.5), \\ \boxed{y} &= H(z_1 - 2z_2 - 0.5). \end{aligned}$$



Difficulty: learning hidden weights, not drawing another layer. For differentiable activations, the chain rule gives credit assignment:

$$\frac{\partial L}{\partial w^{(1)}} = \frac{\partial L}{\partial y} \frac{\partial y}{\partial z} \frac{\partial z}{\partial w^{(1)}}, \quad w \leftarrow w - \eta \nabla_w L.$$

Back-propagation + nonlinear differentiable units + gradient optimization make multilayer learning practical. The hard-threshold network above demonstrates representation, not differentiable training.

Year	Milestone
1969	Minsky–Papert: <i>Perceptrons</i> [1]
1974	Werbos: backward differentiation for learning [3]
1986	Rumelhart–Hinton–Williams: back-propagation milestone [2]

$$\boxed{1986 - 1970 = 16 \text{ years}} \quad (1986 - 1969 = 17).$$

The 1986 work popularized multilayer learning; it was not the first invention of hidden layers. “Dead around 1970” is a simplified description of declining interest.

References: [1] Minsky and Papert, *Perceptrons*, 1969. [2] Rumelhart et al., *Nature* 323, 533–536, 1986. [3] Werbos, *Beyond Regression*, Harvard thesis, 1974 (author’s later account: <https://www.werbos.com/AD2004.pdf>).